Status	Finished
Started	Monday, 19 October 2026, 11:50 AM
Completed	Monday, 19 October 2026, 11:58 AM
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Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Merge Sort
- b. Bubble Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

What is sorting in the context of computer science?

- a. Deleting data from a list
- b. Arranging data in a particular format
- c. Inserting data into a list
- d. Searching for data in a list

The correct answer is: Arranging data in a particular format

Question **3**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By comparing elements sequentially
- b. By sorting data in a single pass
- c. By breaking the input into smaller parts and merging them
- d. By using the bubble-up method

The correct answer is: By breaking the input into smaller parts and merging them

Question **4**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Makes code execution slower
- c. Makes data harder to manage
- d. Decreases the efficiency of algorithms

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **5**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

- a. Linear Search
- b. Merge Sort
- c. Binary Search
- d. Bubble Sort

The correct answer is: Merge Sort

Question **6**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

a. Backtracking
b. Dynamic programming
c. Divide and conquer
d. Greedy method

The correct answer is:
Divide and conquer

Question **7**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

a. Linear Search
b. Binary Search
c. Merge Sort
d. Bubble Sort

The correct answer is: Bubble Sort

Question **8**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Dividing the input into parts, solving each part, and combining the solutions
b. Sorting data in a single pass
c. Sorting data sequentially
d. Rearranging data without sorting

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **9**

Complete

Mark 1.00 out of 1.00

What is one advantage of sorting a list before performing a search operation?

- a. It allows for faster searching
- b. It has no effect on the search operation
- c. It increases the number of comparisons needed
- d. It makes the search operation slower

The correct answer is: It allows for faster searching

Question **10**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It makes the analysis slower
- b. It complicates the analysis process
- c. It allows for quicker identification of duplicates
- d. It has no effect on the analysis process

The correct answer is: It allows for quicker identification of duplicates

Question **11**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Sorting
- b. Searching
- c. Merging
- d. Complexity

The correct answer is:
Complexity

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is known for its simplicity and sequential comparison of elements?

- a. Bubble Sort
- b. Quick Sort
- c. Heap Sort
- d. Merge Sort

The correct answer is: Bubble Sort

Question **13**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of all keys
- b. A sorting algorithm is stable if it preserves the order of non-duplicate keys
- c. A sorting algorithm is stable if it preserves the order of duplicate keys
- d. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **14**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Linear Search
- b. Bubble Sort
- c. Merge Sort
- d. Quick Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

a. Bubble Sort
b. Merge Sort
c. Quick Sort
d. Insertion Sort

The correct answer is: Bubble Sort